

Properties of the Multivariate Normal Distribution

Honours Multivariate Analysis | Lecture 3 Summary | UCT

Property 1 – Linear Combinations Are Normal

- Any single linear combination $\mathbf{a}'\mathbf{X}$ of a MVN vector $\mathbf{X}$ is univariate normal.
- The mean is $\mathbf{a}'\boldsymbol{\mu}$ and the variance is $\mathbf{a}'\boldsymbol{\Sigma}\mathbf{a}$. The converse also holds: if $\mathbf{a}'\mathbf{X}$ is normal for every $\mathbf{a}$, then $\mathbf{X}$ is MVN.

$$\mathbf{a}'\mathbf{X} \sim N(\mathbf{a}'\boldsymbol{\mu}, \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a})$$

Example: Taking $\mathbf{a}' = [1, 0, 0]$ simply picks out $X_1 \sim N(\mu_1, \sigma_{11})$.

Property 2 – Multiple Linear Combinations Are Jointly Normal

- Forming q linear combinations via $\mathbf{Y} = \mathbf{A}\mathbf{X}$ gives a q -dimensional MVN vector.
- The new mean vector is $\mathbf{A}\boldsymbol{\mu}$ and the new covariance matrix is $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'$.

$$\mathbf{A}\mathbf{X} \sim N_q(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$$

Example: In Exercise 3.4, $\mathbf{A}$ is built from $(X_1 + X_2)/2$ and X_3 . Checking $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}'$ is diagonal proves their independence.

Property 3 – All Subsets Are Multivariate Normal

- Every subset of components of a MVN vector is itself MVN with the corresponding sub-mean and sub-covariance block.
- This follows from Property 2 by setting $\mathbf{A} = [\mathbf{I} \mid \mathbf{0}]$.

$$\text{If } \mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \text{then } \mathbf{X}_1 \sim N_q(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$$

Example: If $\mathbf{X} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $(X_1, X_2) \sim N_2(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$ automatically.

Property 4 – Zero Covariance Implies Independence

- For most distributions, zero covariance does **not** imply independence.
- For the MVN it does: if the off-diagonal block $\boldsymbol{\Sigma}_{12} = \mathbf{0}$, the two partitions are fully independent.

$$\mathbf{X}_1 \perp \mathbf{X}_2 \iff \boldsymbol{\Sigma}_{12} = \mathbf{0}$$

Example: With $\Sigma = \begin{pmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, the pair (X_1, X_2) and X_3 are independent since the off-diagonal block is $\mathbf{0}$.

Property 5 – Conditional Distributions Are Normal

- Conditioning on any subset of components preserves normality.
- The conditional mean depends linearly on the conditioning values.
- The conditional covariance is **constant** – it does not depend on the observed value of $\mathbf{X}_2$.

$$\begin{aligned} E(\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2) &= \boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2) \\ \text{Cov}(\mathbf{X}_1 \mid \mathbf{X}_2) &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \end{aligned}$$

Example: This formula was used directly in CA3 Q2 to find the distribution of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$.

Property 6 – Marginal Distributions of Linear Combinations of Independent MVN Vectors

- Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be independent p -dimensional MVN vectors each with the same covariance Σ but possibly different means $\boldsymbol{\mu}_j$.
- Combining them with scalar weights $c_1, \dots, c_n$ gives a new p -dimensional vector $\mathbf{V}_1$ (unlike Property 1, the output is a full vector, not a scalar).
- The covariance uses **squared** weights because the $\mathbf{X}_j$'s are independent, so variances add.

$$\mathbf{V}_1 = \sum_{j=1}^n c_j \mathbf{X}_j \sim N_p \left(\sum_{j=1}^n c_j \boldsymbol{\mu}_j, \left(\sum_{j=1}^n c_j^2 \right) \Sigma \right)$$

- A second combination $\mathbf{V}_2 = \sum b_j \mathbf{X}_j$ is jointly MVN with $\mathbf{V}_1$. They are independent if and only if $\mathbf{b}'\mathbf{c} = 0$.

$$\mathbf{V}_1 \perp \mathbf{V}_2 \iff \mathbf{b}'\mathbf{c} = \sum_{j=1}^n b_j c_j = 0$$

Example: The sample mean $\bar{\mathbf{X}} = \frac{1}{n} \sum_{j=1}^n \mathbf{X}_j$ has all $c_j = 1/n$, giving covariance $\frac{1}{n}\Sigma$.

Example: In Exercise 3.6, $\mathbf{V}_1$ weights $[\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}]$ and $\mathbf{V}_2$ weights $[\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}]$. Since $\mathbf{b}'\mathbf{c} = \frac{1}{16} - \frac{1}{16} - \frac{1}{16} + \frac{1}{16} = 0$, $\mathbf{V}_1$ and $\mathbf{V}_2$ are independent.

Key insight: The MVN is closed under linear transformations, subsetting, and conditioning — all operations remain within the normal family.